

OptiGrid

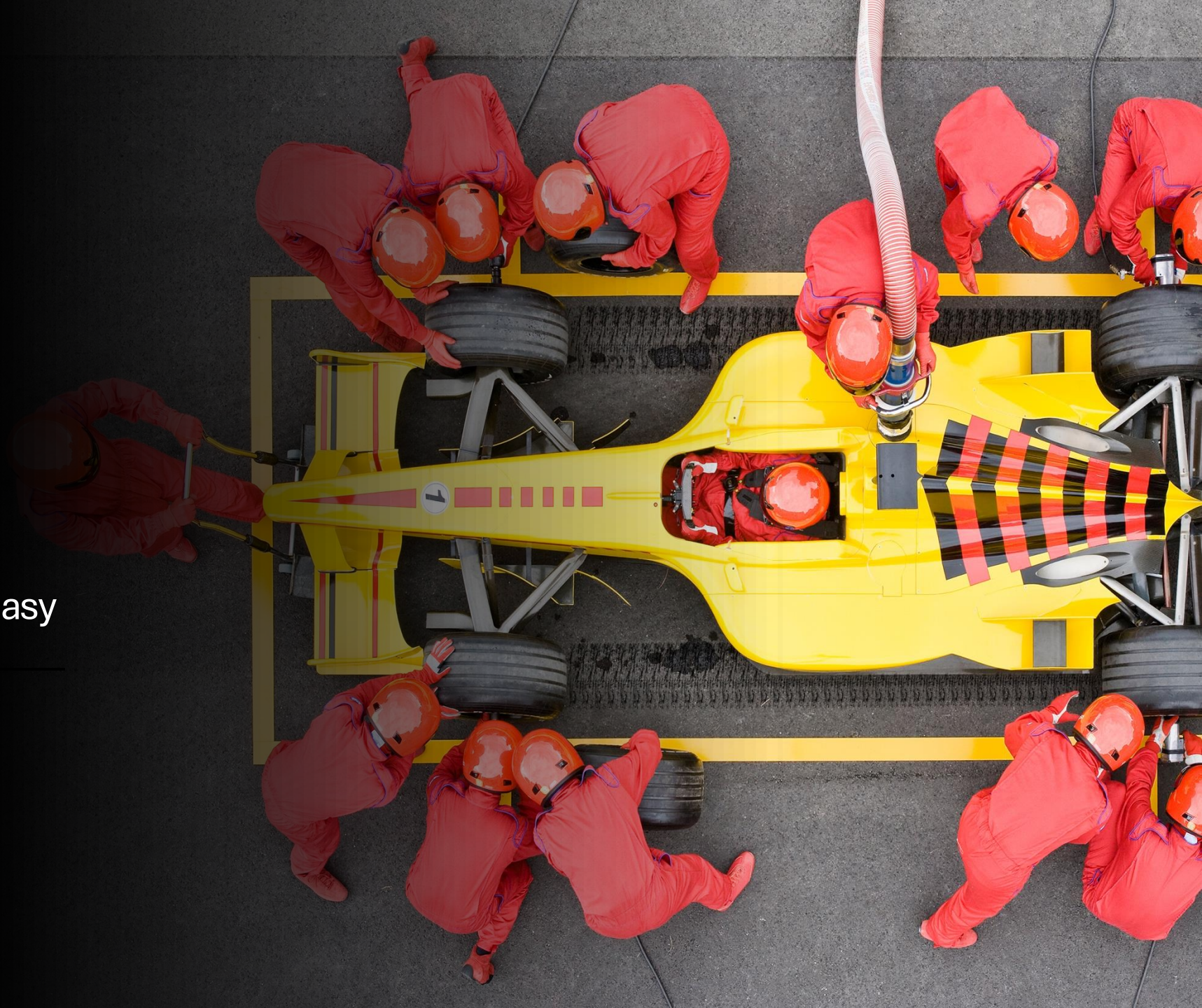
Constrained Optimization and
Visual Risk Profiling for F1 Fantasy

Project in Visualization - 4DT911

Jesper

Emil

Samuel



— PROBLEM

Picking a team under a cost cap

Choose **5 drivers + 2 constructors** within the budget to maximize points, when every race outcome is uncertain.

Constrained

Hard cost cap, fixed roster shape and limited transfers per week.

Uncertain

Crashes, safety cars and weather make averages misleading.

Two objectives

Score points this week and the following weeks

For: F1 Fantasy managers · sports bettors · motorsport analysts and fans

— DATA

Three datasets

HISTORY

Ergast (Kaggle)

6.5M

data points in 14 linked tables: results, qualifying, lap times, pit stops, standings.

MARKET

F1 Fantasy

33 assets

22 drivers and 11 constructors: weekly price, points and ownership since 2022.

CONTEXT

OpenF1 API

2.3M

data points of session weather, sector splits, speed traps and pit flags.

Scope: 2022 onward (Ergast)

Joined on: race_id · driver_id · constructor_id

GOALS

Five Goals

GOAL 1

Integrated Architecture

Unified data pipeline

GOAL 2

Interactive Discovery

Interactive charts and comparisons

GOAL 3

Constrained Optimization

Algorithmic lineup optimizer

GOAL 4

Risk Profiling

Floor (p_{10})

Median (p_{50})

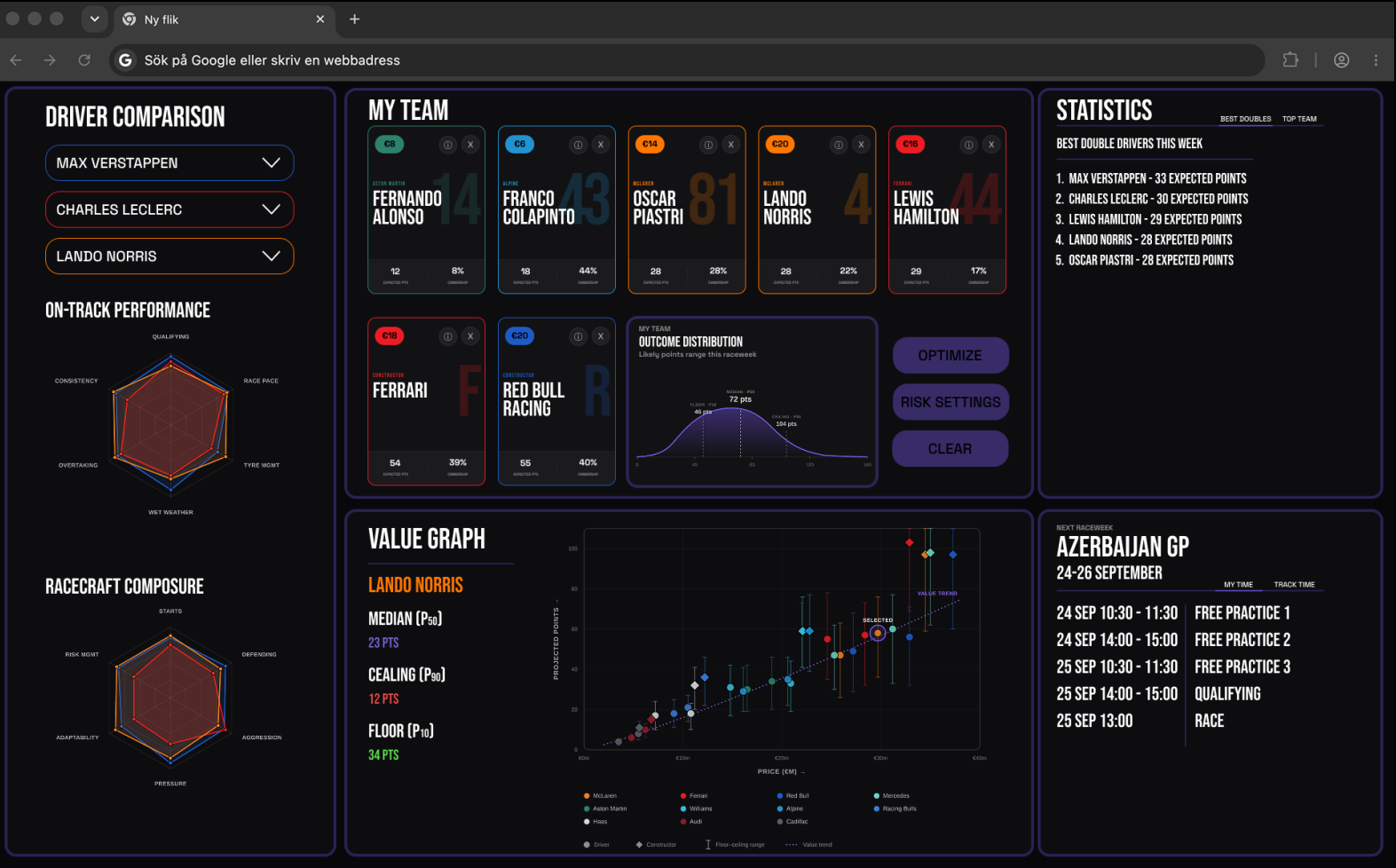
Ceiling (p_{90})

GOAL 5

Dynamic Asset Comparison

Side-by-side comparison

DESIGN



Bubble Chart

PPM comparisons and expected points distribution

Lineup Optimizer

Optimize your lineup from budget and risk slider

Driver Comparison

Comparing of three drivers on different performance metrics

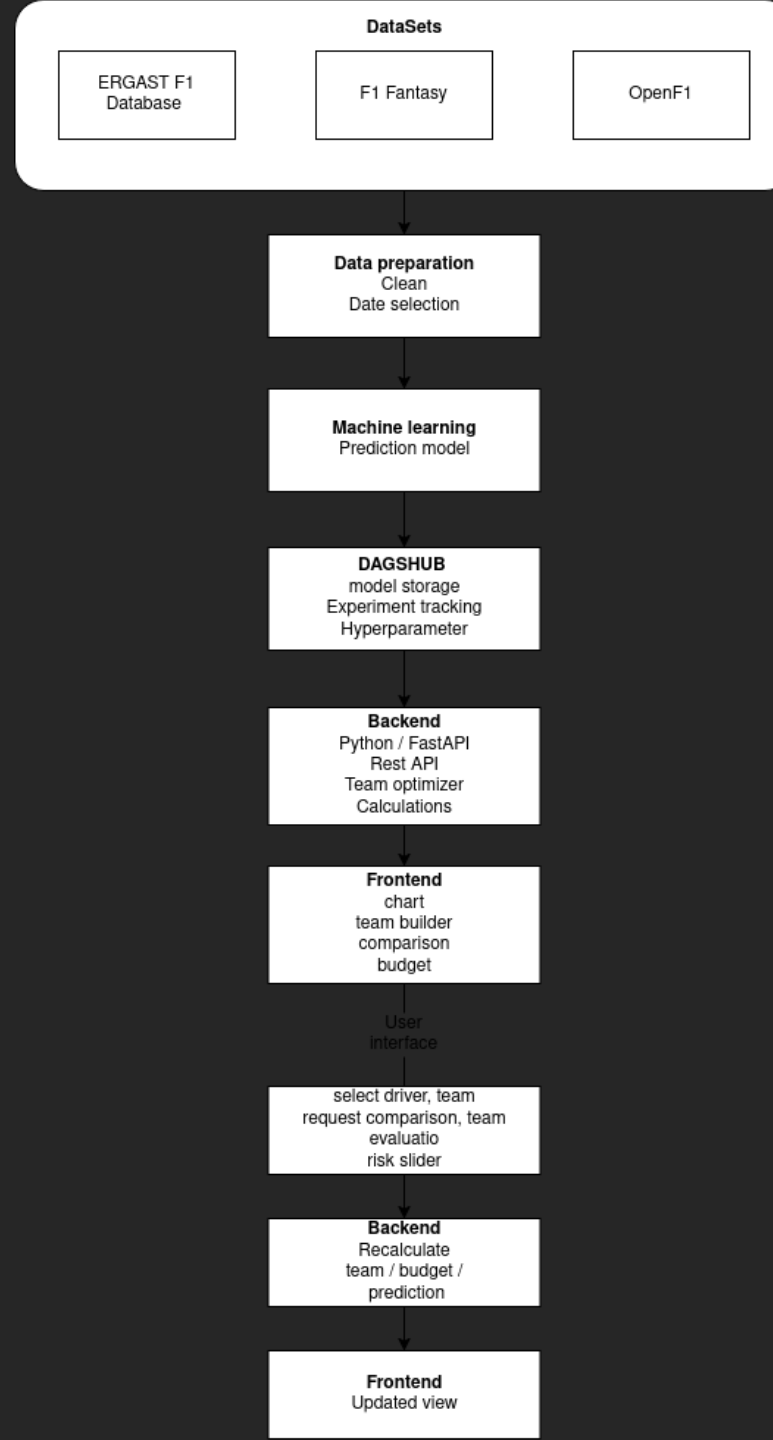
Driver Info

Click on a specific driver to get more information and statistics

Other data

Different statistics and times for upcoming weekend

DATA PIPELINE



— RESPONSIBILITIES

Samuel Berg

DevOps, Backend

Emil Ulvagården

Fullstack

Jesper Wingren

Machine Learning,
Frontend